

Estimating and Predicting Extant Avian Body Mass using Ridge Regression

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Abstract

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1 Introduction and Related Literature

Body mass is important in understanding an organism's lifestyle and interactions with its environment, making it an important biological descriptor (Campione & Evans, 2012), therefore accurate and precise measurements are important, both for extinct and extant organisms, which can be inferred and estimated using skeletal material. There have been many methods in estimating body mass from skeletal material, however in the case of the extant Class Aves (Birds), the femur serves as the primary load bearer given that birds are bipedal, making it a possible predictor of body mass for birds, where the femur serves as the support rather than a pillar. There are many different femoral measurements that can be used to predict body mass such as femoral circumference (Campbell & Marcus, 1992), diameter, or length (Field et al., 2013). According to Campbell and Marcus (1992), the length of the bird's femur has a positive relationship with the bird's body mass, which is explained by how farther the center of gravity is from the point of attachment for the femur to the pelvis and how much longer the femur has to be to extend under it as the bird becomes larger. The same study also discussed that the femur, tibiotarsus and tarsometatarsus reflect body mass in birds, but that the femur in particular has the least variation with regards to general form among different birds, making it a more consistent and reliable indicator of body mass. The femur, as part of the hind limbs, provides total support on land with the expectation that each leg can support the full weight of the animal with little to no support from the forelimbs. Field et al. (2013) made use of thirteen scaling relationships between body mass and skeletal measurements, including different femoral measurements for extant, flying birds. With the use of Ordinary Least Squares regression, their results showed that maximum diameter of the coracoid humerus-articulating facet (HAF) is the most accurate and precise predictor of avian body mass. Sexual variation and differences are also worth considering in mass estimates. Field et al. (2013) found that sex was among the factors that can have a significant impact on the body mass of individual specimens, however Martin-Silverstone et al. (2015) found that among 270 species, 787 total specimens, there is no statistically significant difference in the allometric exponent between the two sexes (58 species of females, 64 species for males) however they concluded that more specimens with identifications on sex are needed to determine significant difference to produce more general results to a wider body size range. In constructing a single, robust model that makes use of multiple skeletal measurements as predictors of mass, Bhavani et al. (2025) made use of Ridge Regression (L2 penalty for Ordinary Least Squares Regression) to mitigate and address multicollinearity since body measurements were expected to be correlated with one another. The same study found that the significant predictors of body weight had a Variance Inflation Factor (VIF) of less than 10. Stability across all body measurements was achieved with the elimination of collinearity using K Ridge Estimators.

2 Dataset and Methods

2.1 Dataset and Univariate Test

The dataset used was sourced from the Vertebrate Zoology collection of the Yale Peabody Museum containing 863 skeletons of extant, flying birds, each with their respective mass with no missing values. The following measurements for each specimen are: maximum femur length (FL), least femur shaft diameter in anterior view (FD), least femur shaft circumference (FC), maximum humerus length (HL), least humerus shaft circumference (HC), least humerus shaft diameter in anterior view (HD), maximum tibiotarsus length (TiL), maximum tarsometatarsus length (TaL), least tarsometatarsus diameter in anterior view (TaD), least tarsometatarsus shaft circumference (TaC), maximum coracoid lateral length (CLL), least coracoid shaft width (CSW), and maximum diameter of the coracoid’s humeral articulation facet (HAF) (Field et al., 2013). Log-Log transformation was necessary for all measurements to account for non-linear relationships and allometric scaling (Pomeroy et al., 2018). Each skeletal measurement underwent univariate linear regression where the coefficient of determination, R^2 , served as the value that indicated the strength of correlation for each skeletal measurement to the body mass of each specimen and goodness of fit.

2.2 Multivariate Test

For equal comparisons between the coefficients of the results, the data was normalized (mean=0, std. dev=1). The multivariate test made use of Ordinary Least Squares Regression with the L2 penalty (Ridge Regression) in analyzing the strength of each predictor, with the L2 penalty in particular being introduced to address multicollinearity resulting in the shrinkage of coefficients. The model is then estimated with the equation:

$$\hat{\beta}^{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

This implementation made use of a train-test split, where 80% of the data served as the training data, while 20% served as the testing set. Predictions were done on the testing set with a 95% confidence interval ($\alpha = 0.05$). All calculations and visualizations were performed in Python 3, with the libraries NumPy, Pandas, Matplotlib, Seaborn, Scikitlearn and StatsModels. The Mathematical framework for this implementation and assumption testing are based on *The Elements of Statistical Learning, Data Mining Inference and Prediction*, Second Edition by Trevor Hastie, Robert Tibshirani and Jerome Friedman and *An Introduction to Statistical Learning with Applications in R, Second Edition* by Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani.

3 Exploratory Data Analysis, Model Implementation and Results

3.1 Exploratory Data Analysis

3.1.1 Importing Libraries, Data Loading and Descriptive Table

```
#Import the libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, RidgeCV
from sklearn.metrics import (mean_squared_error,
mean_absolute_error, r2_score)
from sklearn.model_selection import KFold
import statsmodels.api as sm
import warnings
warnings.filterwarnings('ignore')
np.random.seed(42)

bird_df = pd.read_csv('pone.0082000.s003.csv')
bird_df.describe().round(2).transpose()
```

Output

	mean	std	min	25%	50%	75%	max
mass (g)	619.78	1065.88	2.90	41.25	180	810	9670
fem. circum.	11.68	8.15	1.35	4.85	9.35	16.34	49.15
fem. diam.	3.49	2.44	0.43	1.58	2.89	4.58	14.38
fem. length	40.35	24.16	6.71	20.87	35.57	52.12	155.14
hum. diam.	4.48	3.13	0.58	2.07	3.71	5.85	24.73
hum. length	63.80	52.03	3.44	25.08	53.45	87.84	308.91
tar. circum.	9.98	6.97	1.27	4.19	7.64	14.37	44.19
tar. diam.	3.04	2.11	0.38	1.30	2.50	4.20	13.54
tar. length	45.43	35.52	3.87	22.90	35.81	56.32	332.66
tib. length	70.02	47.74	9.75	36.09	59.93	90.01	399.00
hum. circum.	13.77	9.53	1.82	5.98	11.00	19.17	64.09
cor. HAF	4.70	3.21	0.61	2.10	3.91	6.80	18.51
cor. shaft width	3.38	2.53	0.52	1.47	2.75	4.72	17.23
cor. max length	33.71	19.07	6.23	18.49	28.06	44.96	125.93

Table 1: Descriptive Statistics of Mass and Skeletal Measurements

```
bird_df.info()
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 863 entries, 0 to 862
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Order                                863 non-null    object
1   Genus                                863 non-null    object
2   species                              863 non-null    object
3   Specimen number (YPM)               863 non-null    int64
4   sex                                  863 non-null    object
5   mass (g)                            863 non-null    float64
6   femur_circumference (mm)            863 non-null    float64
7   femur_diameter (mm)                863 non-null    float64
8   femur_length (mm)                  863 non-null    float64
9   humerus_diameter (mm)              863 non-null    float64
10  humerus_length (mm)                863 non-null    float64
11  tarsus_circumference (mm)          863 non-null    float64
12  tarsus_diameter (mm)              863 non-null    float64
13  tarsus_length (mm)                863 non-null    float64
14  tibia_length (mm)                 863 non-null    float64
15  humerus_circumference (mm)         863 non-null    float64
16  coracoid HAF (mm)                 863 non-null    float64
17  coracoid shaft width               863 non-null    float64
18  coracoid max length               863 non-null    float64
dtypes: float64(14), int64(1), object(4)
memory usage: 128.2+ KB
```

3.1.2 Sex Disparity

```
bird_df.sex = bird_df.sex.str.lower()

plt.figure(figsize=(8, 6))
sns.countplot(x='sex', data=bird_df)
plt.title('Distribution by Sex')
print(f'Difference in count: {bird_df.sex.value_counts()[0] -
bird_df.sex.value_counts()[1]}')
print(f'Percentage Difference: {(bird_df.sex.value_counts()[0] -
bird_df.sex.value_counts()[1]) / (bird_df.sex.value_counts()[0] +
bird_df.sex.value_counts()[1]) * 100:.4f}%')
```

Output

Difference in count:73

Percentage Difference: 8.4589%

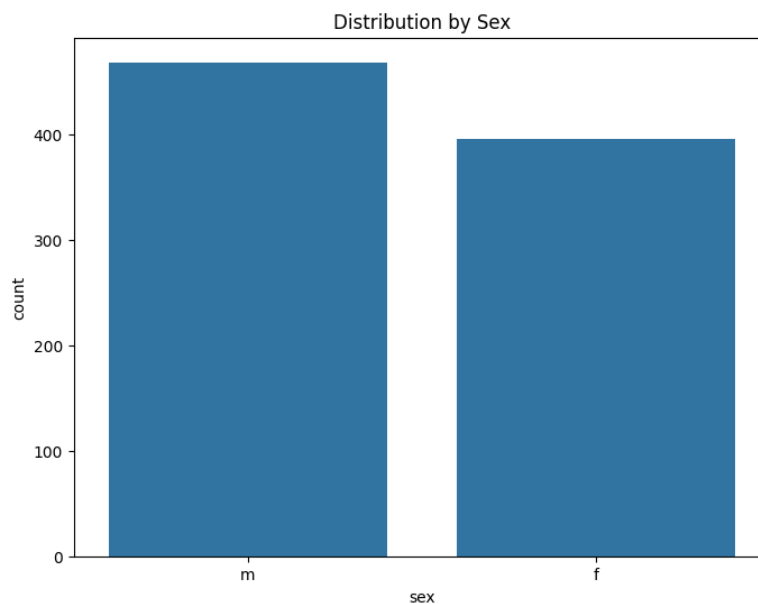


Figure 1: Count graph of Sex

3.1.3 Scatterplot of Measurements to Mass

```
numerical_cols_only_df = bird_df.drop(columns=['Order', 'Genus',
'species', 'Specimen number (YPM)', 'sex', 'mass (g)'])
num_cols_list = numerical_cols_only_df.columns.tolist()

#scatterplot of mass (g) vs. each of the numerical columns
fig, axes = plt.subplots(nrows=4, ncols=4, figsize=(25, 25))
fig.suptitle('Scatterplots of mass (g) vs. Different Skeletal
Measurements', fontsize=16)
axes = axes.flatten()
for i, (col, ax) in enumerate(zip(num_cols_list, axes)):
    sns.scatterplot(x=bird_df[col], y=bird_df['mass (g)'], ax=ax
    hue=bird_df.sex)
    ax.set_title(f'mass (g) vs. {col}')
    ax.set_xlabel(col)
    ax.set_ylabel('mass (g)')
```

Output

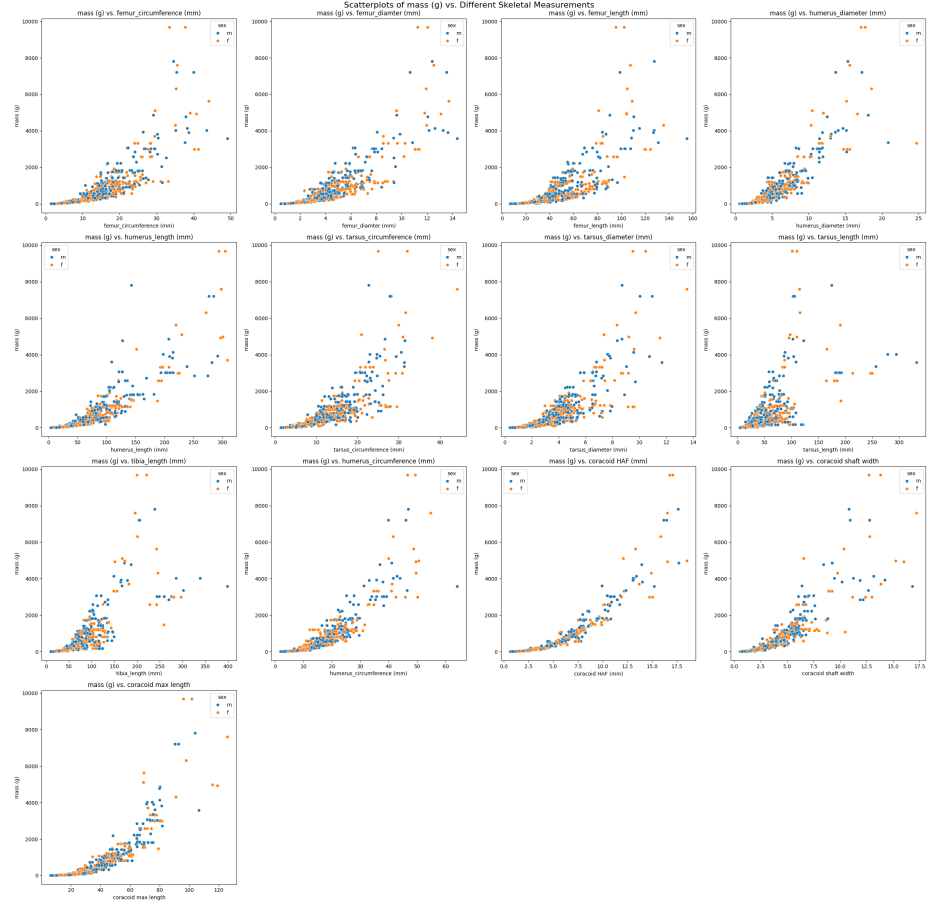


Figure 2: Scatterplots of the Skeletal Measurements and Mass(no transformation)

Figure 2 shows the scatterplot of the skeletal measurements, distinguished in sex by color, where there is a non-linear relationship between the measurements and mass. To account for the non-linear relationship between the skeletal measurements and mass, an assumption needed to perform Ordinary Least Squares Regression, the natural logarithm ($\ln$) of each skeletal measurements and the mass (g) are taken as seen in Figure 3 (below)

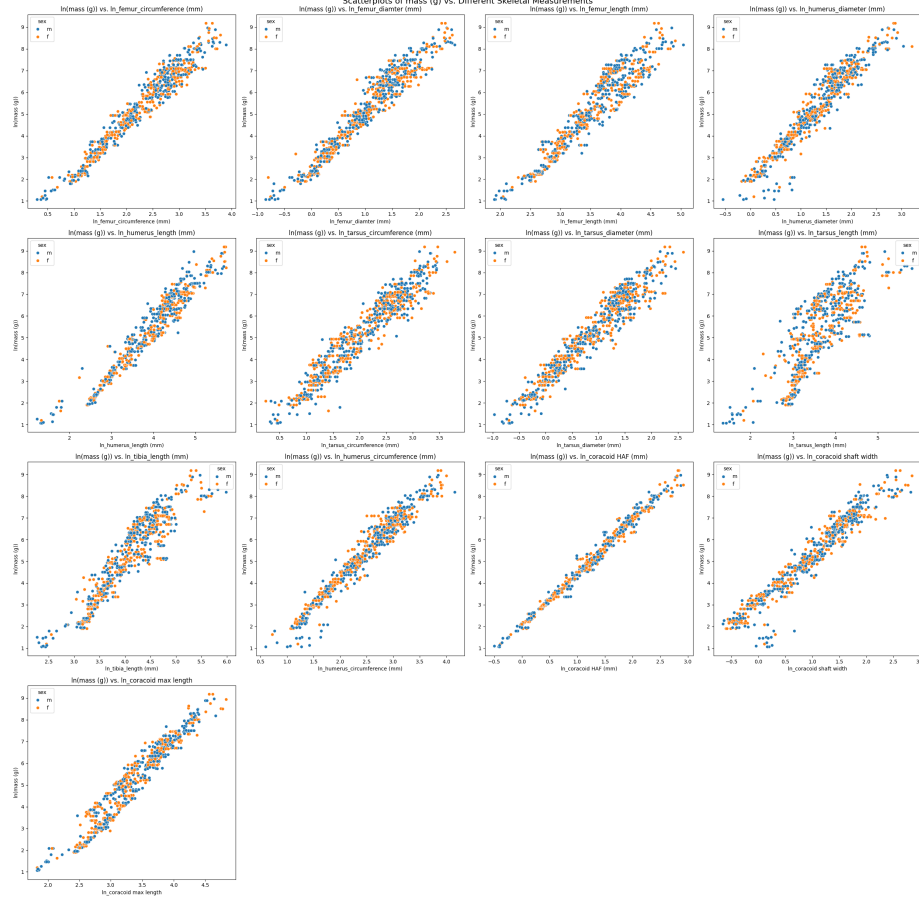


Figure 3: Log Transformation Scatterplots of Mass vs Each Measurement

3.2 Univariate Regression

3.2.1 Linear Regression

```
#using statsmodels, for each measurement
for i in ln_col_names:
    X = sm.add_constant(ln_df[i])
    y = bird_df['mass (g)'].apply(np.log)
    model = sm.OLS(y, X).fit()
    print(f"Results for {i}:")
    print(model.summary())
```

Results

Measurement	R^2
femur circumference (mm)	0.9528
femur diameter (mm)	0.9320
femur length (mm)	0.9029
humerus diameter (mm)	0.9265
humerus length (mm)	0.9364
tarsus circumference (mm)	0.9146
tarsus diameter (mm)	0.9263
tarsus length (mm)	0.6463
tibia length (mm)	0.8580
humerus circumference (mm)	0.9549
coracoid HAF (mm)	0.9881
coracoid shaft width	0.9285
coracoid max length	0.9357

Table 2: Univariate Regression Results

From the results in table 2, the coracoid's humeral articulation facet (HAF) scores the highest R^2 at 0.9881. This is in line with the results of Field et al.(2013) where the length of the coracoid HAF had the highest R^2 score at $R^2 = 0.9881$. Tarsus length had the lowest R^2 value at 0.6463, also in line with the results of Field et al.(2013). The range of values for all R^2 values is (0.6463, 0.9881). The full model summaries are available at the index.

3.2.2 Train-Test Split (Univariate)

```
#using sklearn
lm = LinearRegression()
for i in ln_col_names:
    X = ln_df[[i]]
    y = bird_df['mass (g)'].apply(np.log)
    X_train, X_test, y_train, y_test = train_test_split(X, y,
        test_size=0.2, random_state=42)
    lm.fit(X_train, y_train)
    y_pred = lm.predict(X_test)
    print(f"Results for {i}:")
    print(f" R-squared: {r2_score(y_test, y_pred):.4f}")
    print(f" MSE: {mean_squared_error(y_test, y_pred):.4f}")
    print(f" MAE: {mean_absolute_error(y_test, y_pred):.4f}\n")
```

Results

Skeletal Measurement (mm)	R^2	MSE	MAE
femur_circumference	0.9564	0.1478	0.3056
femur_diameter	0.9422	0.1962	0.3521
femur_length	0.9237	0.2587	0.4194
humerus_diameter	0.9486	0.1745	0.2975
humerus_length	0.9496	0.1712	0.3420
tarsus_circumference	0.9091	0.3084	0.4486
tarsus_diameter	0.9359	0.2176	0.3668
tarsus_length	0.7037	1.0054	0.8871
tibia_length	0.8907	0.3709	0.5113
humerus_circumference	0.9693	0.1040	0.2245
coracoid HAF	0.9906	0.0319	0.1157
coracoid shaft width	0.9604	0.1344	0.2572
coracoid max length	0.9407	0.2012	0.3807

Table 3: Performance Metrics for Univariate Log-Log Model

From the results in table 3, the coracoid's humeral articulation facet (HAF) scores the highest R^2 at 0.9906, while also having the lowest mean squared error (MSE) at 0.0319 or 3.19% and lowest mean absolute error (MAE) at 0.1157 or 11.57%.

3.2.3 Collinearity of Measurements

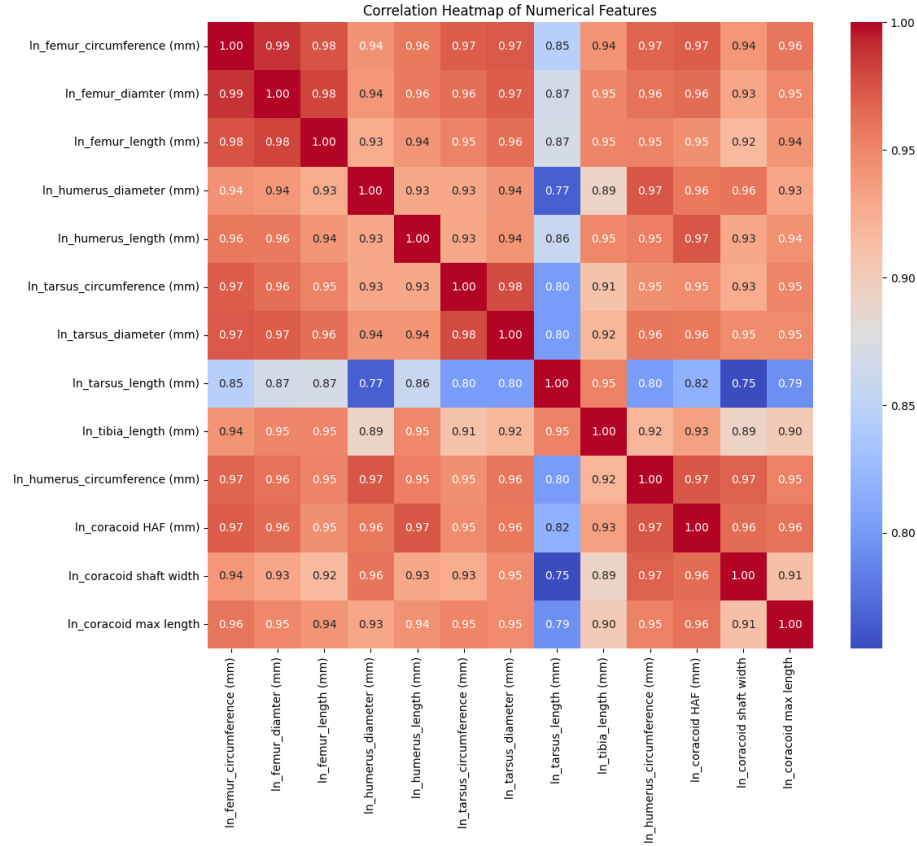


Figure 4: Heatmap Correlation Matrix of the natural logarithm of all measurements

From the heatmap correlation matrix of the natural logarithm of all measurements, all measurements are highly correlated with one another, creating problems when implementing an Ordinary Least Squares Regression using all measurements as predictors, violating an assumption of the model where all predictors must be independent. Further assumption testing is done using a Variance Inflation Factor test to test for Multicollinearity, where the Variance Inflation Factor score (VIF) has a minimum value of 1, which indicates an absence of collinearity. In practice, a VIF score that exceeds 5 or 10 indicates a problematic amount of collinearity (James et al., 2021, p.102). This demonstration made use of this rule of thumb for detecting problematic amounts of collinearity.

```

vif = pd.DataFrame()
vif['VIF'] = [variance_inflation_factor(ln_df.values, i)
for i in range(len(ln_df.columns))]
vif['Variable'] = ln_df.columns
print(vif.sort_values(by='VIF', ascending=False))

```

Output

Variable	VIF
ln_tibia_length (mm)	2378.842806
ln_femur_length (mm)	1127.065158
ln_tarsus_length (mm)	732.806717
ln_coracoid max length	720.860781
ln_femur_circumference (mm)	714.913221
ln_humerus_circumference (mm)	613.978780
ln_humerus_length (mm)	560.964643
ln_tarsus_circumference (mm)	288.583987
ln_coracoid HAF (mm)	204.858344
ln_femur_diameter (mm)	186.012160
ln_humerus_diameter (mm)	105.825774
ln_tarsus_diameter (mm)	95.837043
ln_coracoid shaft width	74.267265

Table 4: Variance Inflation Factor Score for all measurements

Table 4 shows the Variance Inflation Factor score for all measurements, where all measurements exceeded threshold of 5. Hence, the L2 penalty was implemented.

3.3 Multivariate Regression (L2 Linear Regression)

```

kf = KFold(n_splits=5, shuffle=True, random_state=42)
ridge = RidgeCV(alphas=np.logspace(-3,3,100), cv=kf)
X = ln_df
y = bird_df['mass (g)'].apply(np.log)

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
ridge.fit(X_train_scaled, y_train)
y_train_pred = ridge.predict(X_train_scaled)
y_test_pred = ridge.predict(X_test_scaled)

print(f"R-squared: {r2_score(y_test, y_test_pred):.4f}")
print(f"Adjusted R-squared: {1 - (1 - r2_score(y_test,
y_test_pred)) * (len(y_test) - 1) / (len(y_test) -
X_test_scaled.shape[1] - 1):.4f}")
print(f"MSE: {mean_squared_error(y_test, y_test_pred):.4f}")
print(f"MAE: {mean_absolute_error(y_test, y_test_pred):.4f}")
print(f"\nCcoefficients (Standardized):")
for col, coef in zip(ln_df.columns, ridge.coef_):
    print(f" {col}: {coef:.4f}")

```

Results

Measurement	Coefficient	exp(Coefficient)	% Change
ln_femur_circumference	0.2469	1.2801	28.0083
ln_femur_diameter	-0.0356	0.9650	-3.5001
ln_femur_length	0.0888	1.0928	9.2848
ln_humerus_diameter	0.0845	1.0881	8.8134
ln_humerus_length	0.0125	1.0126	1.2596
ln_tarsus_circumference	0.1485	1.1601	16.0084
ln_tarsus_diameter	-0.1903	0.8267	-17.3325
ln_tarsus_length	-0.2030	0.8163	-18.3727
ln_tibia_length	0.1557	1.1685	16.8485
ln_humerus_circumference	-0.0120	0.9881	-1.1944
ln_coracoid HAF	1.2024	3.3281	232.8130
ln_coracoid shaft width	0.0715	1.0742	7.4161
ln_coracoid max length	0.1872	1.2059	20.5913

Table 5: Coefficient Table of Ridge Regression. The % Change column refers to the percent change in mass with respect to a one standard deviation change in the measurement, all others constant

Table 5 shows the coefficient values for all skeletal measurements, and the exponentiated coefficients. The use of a log-log model implies that the results represent the elasticity of mass with respect to a measurement. However, since the measurements were normalized, a change in one standard deviation results in a $[(e^{\beta_i} - 1) \cdot 100]\%$ change in mass, all others constant; where β_i is the coefficient of a measurement. Among the measurements, the maximum diameter of the coracoid humerus-articulating facet (HAF) has the highest percent change in mass at 232.81 % for every increase of one standard deviation, all others constant. Much like in the results of the Univariate test, the maximum diameter of the coracoid humerus-articulating facet (HAF) serves as the greatest predictor of mass among all the measurements. The coefficient shrinkage and cross validation using K-fold for the model is seen in the Ridge Trace plot in Figure 5 (below). The Ridge Trace plot shows the optimal K selected by the K-fold cross validation for the Ridge Regression model, which is at $K \approx 0.0870$.

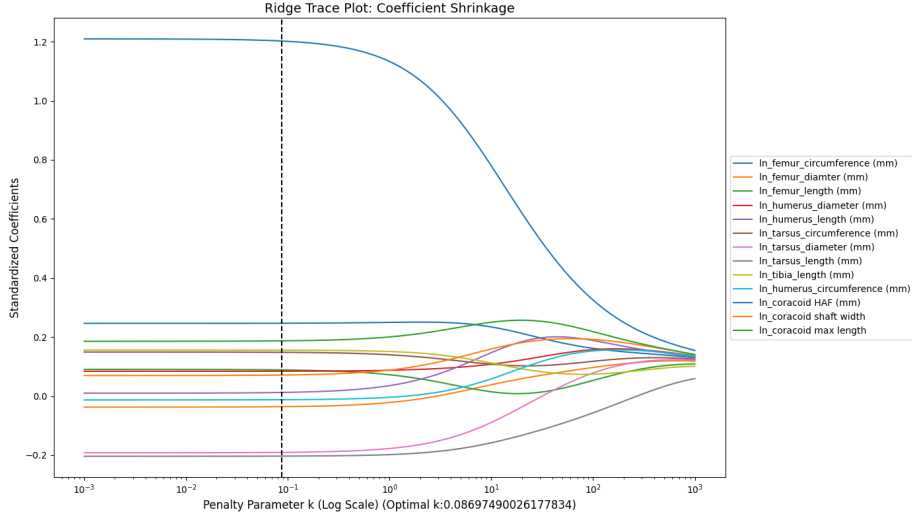


Figure 5: Ridge Trace Plot

Statistic	Value
R^2	0.9942
Adj. R^2	0.9937
Mean Squared. Error	0.0197
Mean Abs. Error	0.0992
Mean Abs. Percentage Error	0.0215

Table 6: Ridge Regression Metrics

Table 6 shows the R^2 , Adjusted R^2 , Mean Standard Error, Mean Absolute Error and Mean Absolute Percentage Error for the Ridge Regression

Model. The R^2 and Adjusted R^2 were above 0.99, which was expected due to the complete measurements in the dataset. The Mean Squared Error is 0.0197, the Mean Absolute Error is 0.0992 while the Mean Absolute Percentage Error is 0.0215. Figure 6 (below) shows the Predicted and Actual values for the natural logarithm of mass with a red diagonal line ($y = x$).

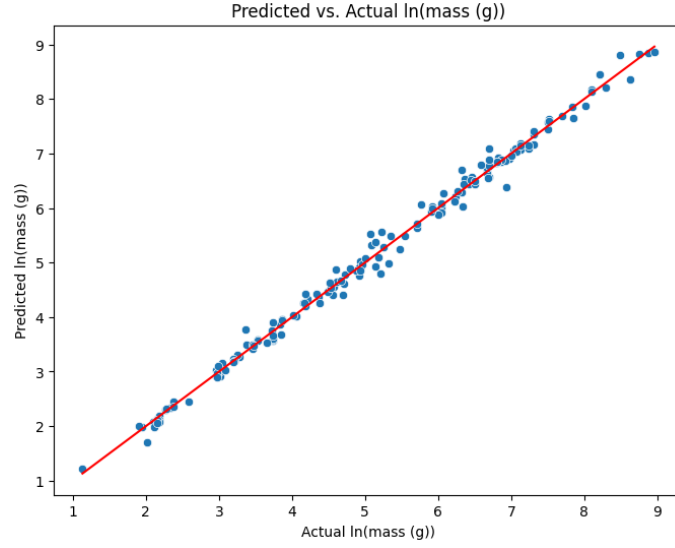


Figure 6: Predicted Vs Actual (Test)

4 Conclusion

Results from the univariate and multivariate tests consistently show that the maximum diameter of the coracoid humerus-articulating facet (HAF) is the best predictor of mass, having the highest R^2 score (0.9906) in the univariate test while having the lowest Mean Squared Error and Mean Absolute Error (0.0319, 0.1157 respectively) and a 232.8130% change in mass for every one increase in standard deviation *ceteris paribus* for the multivariate Ridge Regression test. The L2 penalty for Ridge Regression was tuned using K-fold for cross validation, settling in at $K \approx 0.0870$. Therefore, the maximum diameter of the coracoid humerus-articulating facet (HAF) is the most accurate predictor of Avian body mass.

4.1 Machine Learning Implementation and Implications in Estimating Mass of Extinct Birds

Currently, the Ridge Regression model has been trained on 690 samples, 80% of the 863 samples. Constructing a robust model that predicts the mass of flying birds from skeletal remains requires more samples. Currently, as well, the entire model would need to be re-trained for every set of new entries in the data. Much more work is needed in creating a model that predicts the mass of new entries from incomplete skeletal remains, such is the case for extinct, fossilized remains. Two options would be either to iteratively impute since most of the skeletal remains of this dataset were highly correlated with one another. Another option would be to run multiple sub models where for each sub model, the missing measurement for the new entries are excluded, thereby predicting the mass without the missing measurements.

Disclaimer (AI Declaration)

The author declares that NO generative AI tools were used in synthesizing, writing and editing of this document.

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